

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Error Analysis Task

Course Code:	CSA17	Course Title:	Artificial Intelligence
CO Assessed:	CO1 – Apply (BL3)	Assessment:	Assessment Tool 3 – Error Analysis Task
Weightage:	35% of CIA	Duration / Marks:	60 mins / 100 marks
CO1 Statement:	Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)		
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Section / Batch:	B.Tech-IT	Date:	10/10/2026

Code of Conduct

I Dinesh R-19421117 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Dinesh R

Instructions

- Read each question carefully; answer all four questions (100 in total). Off-Class Assessment Tool.
- Each question gives a student's faulty work. Do not solve from scratch: identify every error (cite it by statement, component, iteration or step number), explain its root cause, correct it with justification and apply the corrected concept. An error merely carried forward counts once.
- Follow the conventions stated in each question (alphabetical expansion, explored set, goal test on removal from the frontier, tie-breaking) and present tables neatly.
- Marks follow the Error Analysis rubric: error identification, root cause analysis, correction and justification, application of concepts, and clarity and presentation.
- Upload in GitHub and GCR (question paper with Reg. No., Name, answers, rubrics and evaluation page at the end).

Section / Topic	Focus	Question Nos.
1. Foundations and History of AI	Four approaches to AI, Turing Test, eras and milestones	Q1
2. Intelligent Agents: Agents and Environments	Agent, percepts, actions, agent function and program, PEAS	Q1
3. The Concept of Rationality	Performance measure, percept sequence, rational vs omniscient	Q1
4. The Nature of Environments	Task-environment properties and their classification	Q1
5. The Structure of Agents	Simple reflex and model-based reflex agent programs	Q1
6. Problem-Solving Agents	Goal formulation, problem formulation, search, execution	Q2
7. Problem Formulation	States, initial state, actions, goal test, path cost; belief states	Q2, Q4
8. Searching for Solutions	Search graph, frontier, explored set, node expansion	Q3

9. Uninformed Search Strategies	Error tracing of BFS, DFS and uniform-cost search	Q3
10. Search with Partial Information	Sensorless problems, belief states, percept update	Q4

Annexure A – Error Analysis Task

Question 1: Foundations of AI, Intelligent Agents and Rationality – Error Analysis of Concept Notes

Topic: Foundations and history of AI, agents and environments, rationality, nature of environments, structure of agents

A junior engineer prepared the revision notes below on the foundations of AI and intelligent agents. The notes contain errors; some statements may be correct. Analyse them statement by statement.

No.	Statement in the notes
S1	The Turing Test (1950) judges a machine to be intelligent if its internal reasoning always follows the laws of logic. It is therefore the standard example of the “thinking rationally” approach to AI.
S2	The “AI winters”, periods of reduced funding and interest, occurred in the 1950s, before the 1956 Dartmouth workshop. That workshop then revived the field and coined the term “artificial intelligence”.
S3	A rational agent always achieves the best possible outcome because it knows the actual result of every action in advance. Hence a taxi that picks a route later blocked by an unforeseeable landslide was not rational.
S4	PEAS description of an automated taxi – Performance measure: steering, accelerator, brake, horn. Environment: roads, other traffic, pedestrians, weather. Actuators: safe, fast, legal and comfortable trip. Sensors: cameras, GPS, speedometer.
S5	Chess played without a clock is a partially observable, stochastic, episodic, static and discrete task environment.
S6	A model-based reflex agent keeps an internal state, updated using knowledge of how the world evolves and how its own actions affect it. It can therefore act sensibly even when the current percept does not reveal the full state of the world.

a) Correct or incorrect statements

Statement	Status	Erroneous phrase / claim
S1	Incorrect	The Turing Test tests whether a machine behaves humanly, not whether its internal reasoning always follows logic. It is “acting humanly,” not “thinking rationally.”
S2	Incorrect	AI winters did not occur in the 1950s before Dartmouth. The 1956 Dartmouth workshop helped establish AI and coined the term; AI winters came later.
S3	Incorrect	A rational agent does not know every action’s outcome in advance and need not achieve the best possible outcome.
S4	Incorrect	Performance measure and actuators are swapped. Steering, accelerator, brake and horn are actuators; safe, fast, legal and comfortable trip are performance criteria.
S5	Incorrect	Chess without a clock is not partially observable, stochastic or episodic. The board and pieces are visible, moves are rule-based, and moves affect later states.
S6	Correct	No change. A model-based reflex agent maintains internal state and uses a

		model of how the world and its actions change the state.
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(b) Root cause analysis

Error	Root cause / misconception
S1	Confuses acting humanly with thinking rationally. The Turing Test evaluates observable conversational behaviour.
S2	Confuses the start of AI with later periods of reduced funding and interest. AI winters occurred after the field had developed.
S3	Confuses rationality with omniscience. Rational decisions use percepts, knowledge, available actions and expected performance under uncertainty.
S4	Does not understand the PEAS categories: performance measure describes success; actuators are means of acting on the environment.
S5	Misclassifies the chess environment. Board state is observable, legal moves are deterministic, and the game is sequential.

(c) Corrected statements and justifications

Statement	Corrected statement	Justification
S1	The Turing Test (1950) evaluates whether a machine can behave indistinguishably from a human in conversation. It represents acting humanly.	It evaluates external behaviour, not whether internal reasoning follows formal logic.
S2	The 1956 Dartmouth workshop helped launch AI and introduced the term “artificial intelligence.” AI winters occurred later, during periods of reduced funding and interest.	The chronology places AI winters after the field had developed.
S3	A rational agent chooses the action expected to maximize its performance measure using available percepts and knowledge. An unforeseeable landslide does not by itself make the taxi irrational.	Rationality is judged by information available at decision time, not by hindsight.
S4	Performance measure: safe, fast, legal and comfortable trip. Environment: roads, traffic, pedestrians and weather. Actuators: steering, accelerator, brake and horn. Sensors: cameras, GPS and speedometer.	Each PEAS item is placed in the correct category.
S5	Chess without a clock is fully observable, deterministic, sequential, static and discrete; it is also multi-agent.	The board is visible, moves follow rules, moves affect future states, and no clock makes the environment static.
S6	No change. A model-based reflex agent stores an internal state updated using its model of the world and its actions.	This supports sensible action when a single percept does not show the complete state.

(d) Apply corrected concepts to the taxi case

(i) Was the route choice rational? Yes, based on the facts given. The taxi used the available traffic reports, which showed the highway clear, and selected a route expected to perform well. The later unforeseeable landslide does not make the earlier decision irrational.

(ii) Perfect foresight is impossible because the taxi environment is partially observable, dynamic and stochastic. Conditions can change unexpectedly and sensors/reports cannot reveal every future event.

(iii) A simple reflex taxi may continue toward a road that has become blocked because its current rule does not remember earlier information or represent the changing road situation. A model-based reflex taxi keeps an internal state/model of roads, traffic conditions and the effects of actions, updating it as new percepts arrive.

Question 2: Problem-Solving Agents and Problem Formulation – Error Analysis of a Formulation

Topic: Problem-solving agents, problem formulation

A student formulates the **Missionaries and Cannibals** problem for a problem-solving agent. Three missionaries and three cannibals stand on the left bank of a river, with a boat that carries one or two people and cannot cross empty. The boat is on the left bank at the start. Missionaries must never be outnumbered by cannibals on either bank (a bank with no missionaries is safe). The objective is to move everyone to the right bank using the **fewest crossings**.

Component	Student's formulation
State representation	(M, C) = number of missionaries and cannibals on the left bank
Initial state	(3, 3)
Actions	Take the boat across with 1M, 2M, 1C, 2C or 1M + 1C
Goal test	$M = C$
Legality check	A state is legal if $C \leq M$ on the left bank (or $M = 0$ there)
Path cost	Number of people carried in the crossing
Note N1	The agent works in the order: search $\rightarrow$ goal formulation $\rightarrow$ problem formulation $\rightarrow$ execution.
Note N2	The state space contains 16 states, of which 10 are legal.

(a) Check each component and note

Component / note	Status	What is wrong
State representation	Incorrect / incomplete	(M,C) omits the boat location. Use (M,C,B), where B is L or R.
Initial state	Incomplete	Use (3,3,L), including that the boat starts on the left.
Actions	Mostly correct but incomplete	These loads are possible, but a crossing is allowed only if the people are available on the boat's current bank and the resulting state is legal.
Goal test	Incorrect	$M=C$ is not the goal. Goal is $M=0$ and $C=0$ on the left (everyone is on the right).
Legality check	Incorrect	It checks only the left bank. Both banks must be safe: on each bank, missionaries must be at least as many as cannibals unless there are zero missionaries.
Path cost	Incorrect	The objective is fewest crossings, so each crossing costs 1, not the number of people carried.
Note N1	Incorrect	Correct order: goal formulation $\rightarrow$ problem formulation $\rightarrow$ search $\rightarrow$ execution.
Note N2	Incorrect	With M, C each ranging 0–3 and boat side L/R, there are $4 \times 4 \times 2 = 32$ possible encoded states. 20 satisfy the safety constraints.

(b) Root cause analysis

Error	Root cause
State representation / initial state	The state must contain all information needed to predict actions, including the boat side.
Actions	Actions must respect boat capacity, boat location, people available and legal resulting states.
Goal test	The goal must represent all people reaching the right bank, not merely equal counts on the left.
Legality check	Safety constraints apply to both banks, not only the left bank.
Path cost	The problem explicitly minimizes crossings, so each crossing has equal cost 1.
N1	The agent must decide the goal and define the problem before searching and executing a plan.
N2	The boat side doubles the number of possible encoded states; the student omitted it.

(c) Corrected formulation

Component / note	Corrected formulation	Justification
State	(M,C,B): missionaries and cannibals remaining on left, plus boat side $B \in \{L,R\}$.	Includes all information required to determine legal crossings.
Initial state	(3,3,L).	All six people and the boat start on the left.
Actions	Move 1M, 2M, 1C, 2C, or 1M+1C across, if available on the boat's bank; reject illegal results.	Respects capacity and safety.
Goal test	(0,0,R) in the state convention, or M=0 and C=0 on the left with everyone on right.	Ensures all people have crossed.
Legality	On each bank, missionaries are not outnumbered by cannibals unless that bank has zero missionaries.	Checks both banks.
Path cost	1 per crossing; minimize total crossings.	Matches the stated objective.
N1	Goal formulation → problem formulation → search → execution.	Correct problem-solving-agent sequence.
N2	32 possible encoded states; 20 safe/legal states when boat side is included.	There are 16 M,C pairs and two boat locations; 10 pairs are safe, each with two boat-side values.

(d) Apply the corrected formulation

(i) Loads available from the initial state

Boat load	Resulting state (M,C,boat)	Legal? Reason
1M	(2,3,R)	No. Left bank has 2 missionaries and 3 cannibals; missionaries are outnumbered.
2M	(1,3,R)	No. Left bank has 1 missionary and 3 cannibals.
1C	(3,2,R)	Yes. Left bank is safe (3M,2C); right bank has no missionaries.
2C	(3,1,R)	Yes. Left bank is safe (3M,1C); right bank has no missionaries.
1M + 1C	(2,2,R)	Yes. Left bank has equal numbers; right bank has no missionaries.

(ii) Example accepted by the student but actually illegal: (M,C,B) = (2,1,L). The left bank passes $C \leq M$, but the right bank has 1 missionary and 2 cannibals, so it is unsafe.

(iii) The optimal solution costs 11 crossings when each crossing costs 1. This is the standard minimum-crossing solution for the 3-missionary, 3-cannibal problem.

Question 3: Searching for Solutions and Uninformed Search – Error Analysis of Search Traces

Topic: Searching for solutions, BFS, DFS, uniform-cost search

A student ran BFS, DFS and uniform-cost search (UCS) on the weighted undirected graph below. The start node is S and the goal node is G. **Conventions:** graph search with an explored set; neighbours expanded alphabetically; goal test when a node leaves the frontier; ties broken alphabetically. BFS and DFS ignore edge costs while searching, but path costs are measured with them. **BFS:** do not add a node that is already explored or already in the queue. **DFS:** use a stack with its top on the left; push children so that the alphabetically first child is popped first; do not push a node that is already explored or already on the stack. **UCS:** write entries as Node:g(parent) and replace an entry whenever a cheaper path to a frontier node is found.

Edge	Cost	Edge	Cost
S – C	3	B – G	3
S – D	3	D – E	2
A – B	2	D – G	6
A – C	6	E – G	3
B – D	4		

Student's BFS trace (queue initially [S])

Iteration	Node Dequeued	Queue after Expansion	Explored Set
1	S	[C, D]	{S}
2	C	[D, A]	{S, C}
3	D	[B, E, G, A]	{S, C, D}
4	B	[E, G, A]	{S, C, D, B}
5	E	[G, A]	{S, C, D, B, E}
6	G (goal)	[A]	{S, C, D, B, E}

BFS result: path S – D – G, cost 9, 6 nodes removed from the frontier.

Student's DFS trace (stack initially [S], top on the left)

Iteration	Node Popped	Stack after Pushing Children
1	S	[D, C]
2	D	[G, E, B, C]
3	G (goal)	[E, B, C]

DFS result: path S – D – G, cost 9, 3 nodes removed from the frontier.

Student's UCS trace

Iteration	Node Expanded	g(n)	Priority Queue after Expansion
1	S	0	C:3(S), D:3(S)
2	C	3	D:3(S), A:9(C)
3	D	3	E:5(D), B:7(D), A:9(C), G:9(D)
4	E	5	B:7(D), A:9(C), G:9(D)
5	B	7	A:9(C), G:9(D)
6	A	9	G:9(D)
7	G (goal)	9	–

UCS result: path S – D – G, cost 9, 7 nodes removed from the frontier.

Student's remarks

No.	Remark
R1	An explored set is not needed in BFS because BFS never revisits a node, so the Explored Set column is unnecessary.
R2	All three strategies returned S – D – G with cost 9, so all three are optimal and the traces are confirmed correct.
R3	DFS removed the fewest nodes from the frontier (3), so DFS is the best strategy for this graph.

a) Errors in the given traces and remarks

No.	Where	What is wrong
1	BFS iteration 3	A was already in the queue from iteration 2. After expanding D, it must remain at the front; D's neighbours are B, E, G and S, not A. Correct queue is [A,B,E,G].
2	BFS iterations 4–6	The later dequeue order is wrong because the queue at iteration 3 was wrong. Correct order continues A, B, E, then G.
3	DFS iteration 1	The stack should be [C,D], not [D,C], because the alphabetically first child must be popped first.
4	DFS result	The given path S–D–G follows the incorrect stack order. Correct DFS reaches G via S–C–A–B–G.
5	UCS iteration 4	Expanding E finds G at cost 5+3=8, cheaper than the existing G cost 9. The frontier entry for G must be replaced.
6	UCS iterations 5–7 and result	These rows carry forward the failure to update G. Correct UCS removes G at cost 8 after B, with path S–D–E–G.
7	Remark R1	An explored set is needed in graph search to avoid repeated expansion and cycles, including in BFS.
8	Remark R2	The strategies do not all return the same path or cost; BFS and DFS are not generally cost-optimal. UCS finds the cost-8 optimal path here.
9	Remark R3	Fewest nodes removed does not make DFS the best strategy. Path quality, completeness and optimality also matter.

(b) Root cause analysis

Error	Root cause
BFS queue	Did not preserve the existing frontier order and incorrectly added a node that was not a neighbour of D.
DFS stack	Did not follow the stated stack convention for alphabetical-first popping.
UCS queue	Failed to relax/update a frontier node when a cheaper route was found.
R1	Confuses tree search with graph search; undirected edges can lead back to explored nodes.
R2 / R3	Confuses reaching the goal quickly with finding a minimum-cost solution. BFS/DFS ignore edge costs; UCS uses them.

(c)(i) Corrected BFS trace

Iteration	Node dequeued	Queue after expansion	Explored set
1	S	[C,D]	{S}
2	C	[D,A]	{S,C}
3	D	[A,B,E,G]	{S,C,D}
4	A	[B,E,G]	{S,C,D,A}
5	B	[E,G]	{S,C,D,A,B}
6	E	[G]	{S,C,D,A,B,E}
7	G (goal)	[]	{S,C,D,A,B,E,G}

Correct BFS dequeue order: S, C, D, A, B, E, G. Queue after iteration 3: [A, B, E, G]. The BFS path is S–D–G, cost 3+6=9, with 7 nodes removed from the frontier.

(c)(ii) Corrected DFS trace

Iteration	Node popped	Stack after pushing children (top left)
1	S	[C,D]
2	C	[A,D]
3	A	[B,D]
4	B	[G,D]
5	G (goal)	[D]

Correct DFS path: S–C–A–B–G. Cost = 3+6+2+3 = 14. Nodes removed from frontier = 5.

(c)(iii) Corrected UCS trace

Iteration	Node expanded	g(n)	Priority queue after expansion (lowest cost first)
1	S	0	C:3(S), D:3(S)
2	C	3	D:3(S), A:9(C)
3	D	3	E:5(D), B:7(D), A:9(C), G:9(D)
4	E	5	B:7(D), G:8(E), A:9(C)
5	B	7	G:8(E), A:9(C)
6	G (goal)	8	—

UCS updates G from cost 9 to cost 8 after expanding E. Correct path: S–D–E–G, cost 3+2+3=8. Nodes removed from frontier = 6.

Justification: BFS expands by depth, DFS follows the stack order, and UCS always expands the frontier node with the lowest accumulated path cost. UCS must replace a frontier entry whenever a cheaper route is found.

(d)(i) Comparison

Strategy	Path found	Path cost	Nodes removed	Optimal here?
BFS	S–D–G	9	7	No; not minimum cost
DFS	S–C–A–B–G	14	5	No
UCS	S–D–E–G	8	6	Yes

(d)(ii) Why is an explored set needed? At BFS iteration 3, D has a connection back to S, which has already been explored. The explored set prevents S from being added and expanded repeatedly, avoiding cycles and unnecessary work.

Question 4: Search with Partial Information – Error Analysis of Belief-State Computation

Topic: Problem formulation, search with partial information (sensorless and percept-based belief-state search)

A vacuum agent lives in a two-square world (Left square, Right square), each square being Clean (C) or Dirty (D). Actions: Left, Right and Suck (Left in the left square and Right in the right square leave the agent where it is; Suck cleans the current square). The goal is that both squares are Clean. Physical states are written (Agent, Left square, Right square):

State	Description	State	Description
S1	(L, D, D)	S5	(L, C, D)
S2	(R, D, D)	S6	(R, C, D)
S3	(L, D, C)	S7	(L, C, C)
S4	(R, D, C)	S8	(R, C, C)

Task A – sensorless agent. The agent knows it is in the Left square but nothing about the dirt, so its initial belief state is {S1, S3, S5, S7}. A student applied the plan [Suck, Right, Suck]:

Step	Action	Belief State after Action
0	(start)	{S1, S3, S5, S7}
1	Suck	{S5, S7}
2	Right	{S6, S8}
3	Suck	{S6, S8}

Task B – agent with a local sensor that reports [location, status of the current square]. Starting from the belief state of all eight states, the agent receives the percept [Left, Dirty], performs Suck, receives [Left, Clean], performs Right and receives [Right, Dirty]. The student's table:

Step	Action	After Action (Predicted)	Percept	After Percept (Updated)
0	(start)	{S1, S2, S3, S4, S5, S6, S7, S8}	[Left, Dirty]	{S1, S2, S3, S6}
1	Suck	{S4, S5, S7, S8}	[Left, Clean]	{S5, S7}
2	Right	{S6, S7}	[Right, Dirty]	{S6}

Student's conclusions

No.	Conclusion
C1	Task A: the final belief state still contains two states, so the agent can never know where it is. No plan can guarantee the goal when the agent has no sensors; hence sensorless problems cannot be solved.
C2	Task B: the belief state is now the singleton {S6}, so the agent knows its location and the dirt status, and the goal has been reached.
C3	With 8 physical states there are only 8 belief states (one per physical state), so searching over belief states is no harder than searching over physical states.

a) Errors in tables and conclusions

No.	Where	What is wrong
1	Task A, step 3 (Suck)	After Right, belief state is {S6,S8}. Suck cleans the right square when dirty, so both states transition to S8. The result should be {S8}, not {S6,S8}.
2	Task B, step 0 update	Percept [Left, Dirty] means the agent is at L and the left square is dirty. Updated belief state should be {S1,S3}, not {S1,S2,S3,S6}.
3	Task B, step 1 prediction	After Suck from {S1,S3}, predicted state is {S5,S7}, not {S4,S5,S7,S8}.
4	Task B, step 2 prediction	After Right from {S5,S7}, predicted state is {S6,S8}, not {S6,S7}.
5	Conclusion C1	A singleton belief state is not required to guarantee a goal. The corrected sensorless plan ends at {S8}, which is a goal state.
6	Conclusion C2	{S6} does not satisfy the goal: S6=(R,C,D), so the right square is dirty. Knowing the exact state does not mean the goal is reached.
7	Conclusion C3	Eight physical states produce $2^8=256$ possible belief states (including the empty set), not only eight.

(b) Root cause analysis

No.	Root cause
1	The Suck transition was not applied to every physical state in the belief set.
2	The percept was not used to filter states by both location and current-square status.
3	The action prediction did not consistently apply Suck to each possible state.
4	The Right action changes the agent location to R but does not change dirt status; S7 must become S8.
5	Confuses knowing the exact physical state with being able to achieve a goal; a sensorless plan can guarantee a goal across possibilities.
6	Confuses a singleton belief state with a goal state. Goal requires both squares clean.
7	Confuses physical states with subsets of physical states. Every subset can be a belief state.

(c)(i) Corrected Task A belief states

Step	Action	Belief state after action
0	Start	{S1,S3,S5,S7}

1	Suck	{S5,S7}
2	Right	{S6,S8}
3	Suck	{S8}

The plan [Suck, Right, Suck] is a solution: the final belief state is {S8}, where the agent is at R and both squares are clean.

(c)(ii) Corrected Task B prediction and update

Step	Action	After action (predicted)	Percept	After percept (updated)
0	Start	{S1,S2,S3,S4,S5,S6,S7,S8}	[Left, Dirty]	{S1,S3}
1	Suck	{S5,S7}	[Left, Clean]	{S5,S7}
2	Right	{S6,S8}	[Right, Dirty]	{S6}

(c)(iii) Corrected conclusions

C1: Incorrect. The sensorless plan reaches {S8}, where both squares are clean. The agent does not need to know its exact state if every possible state in its belief set satisfies the goal.

C2: Incorrect. The belief state {S6} identifies the agent at R with left clean and right dirty. The goal has not been reached because the right square remains dirty.

(d)(i) Number of belief states

For 8 physical states, the number of subsets is $2^8 = 256$ possible belief states, including the empty set. Belief-state search is harder because it may search up to 256 belief states instead of only 8 physical states.

(d)(ii) Apply plan [Right, Suck] to {S1,S3,S5,S7}

Step	Action	Belief state after action
0	Start	{S1,S3,S5,S7}
1	Right	{S2,S4,S6,S8}
2	Suck	{S4,S8}

Is it a solution? No. S4=(R,D,C) still has the left square dirty, so not every state in the final belief state satisfies the goal.

Rubrics for Error Analysis

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Error Identification	20	Accurately identifies all errors with clear understanding of issues	Identifies most errors with minor omissions	Identifies limited errors; some are missed	Fails to identify key errors
Root Cause Analysis	30	Clearly explains underlying causes with strong technical reasoning	Explains causes with moderate clarity	Partial explanation; weak reasoning	Unable to identify causes of errors
Correction & Justification	25	Provides correct solutions with strong justification and logic	Provides correct corrections with some justification	Corrections are partially correct or weakly justified	Incorrect or no corrections provided
Application of Concepts	15	Effectively applies relevant concepts to analyze and resolve errors	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts
Clarity & Presentation	10	Presents analysis clearly, logically, and systematically	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q. No. / Criteria	Error Identification	Root Cause Analysis	Correction & Justification	Application of Concepts	Clarity & Presentation	Sub. Total	Total %
1							
2							
3							
4							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____